Name: ______________________

Exit quiz

Problem solving with transformations

1 The perimeter and area are the same after a translation, rotation or reflection because the object and image are ____________________ . Fill in the blank

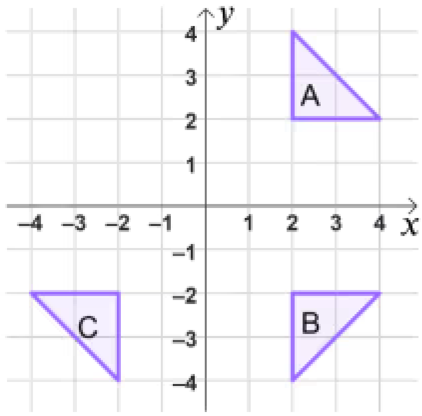
2 A shape is rotated 90° anti-clockwise about $(2, 7)$. Which of the following rotations would result in the image being moved to the same position as the original object? Tick **2** correct answers

- [ ] Rotation 90° clockwise about $(2, 7)$
- [ ] Rotation 90° clockwise about $(7, 2)$
- [ ] Rotation 270° anti-clockwise about $(2, 7)$

3 The coordinate (a, b) has been translated by $\begin{pmatrix}4\\3\end{pmatrix}$. The image of this point will have the coordinate __________. Tick **1** correct answer

- [ ] $(4a, 3b)$
- [ ] $(a4, b3)$
- [ ] $(a + 4, b + 3)$
- [ ] $(a - 4, b - 3)$

4 Shape A is reflected in the x-axis to create shape B. Shape B is then reflected in the y-axis to create shape C. Which transformations would map shape C onto shape A? Tick **2** correct answers



- ☐ A rotation 180° about the origin.
- ☐ A reflection in the line that passes through $(-4, 4)$ and $(3, -3)$
- ☐ A translation by $\begin{pmatrix} 4 \\ 4 \end{pmatrix}$
- ☐ A reflection in the line that passes through (4, 4) and (-2, -2)

5 A shape with perimeter 16 cm is enlarged by scale factor 3. The perimeter of the image is ____________________ cm. Fill in the blank

6 The point A(p + 3, q + 5) has the image A'(p + 1, q + 8). Which vector has it been translated by? Tick **1** correct answer

- ☐ $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$
- ☐ $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$
- ☐ $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$
- ☐ $\begin{pmatrix} -2p \\ 3q \end{pmatrix}$